



PROGRESS REPORT

Title:

Development and Characterization of Bio-Composites from Waste Aquatic Biomass for Sustainable Insulation.

Introduction:

The project aims to develop sustainable insulation materials using waste aquatic biomass, primarily *water Lily*. The research focuses on utilizing its fibrous structure to form bio-composite panels and evaluating their physical, mechanical, and thermal performance in comparison with conventional insulating materials. This project supports the circular economy approach by transforming aquatic waste into functional insulation materials that are biodegradable and recyclable. Furthermore, at the end of their service life, these bio-composites can be thermochemically processed (e.g., through pyrolysis) to recover useful energy, thereby minimizing waste and creating an additional sustainability benefit.

Problem Statement:

Synthetic petroleum-based insulators as well as waste aquatic biomass are posing a threat to environment. Our goal is to convert waste aquatic biomass into bio composite insulation panels offering an environment friendly alternative to traditional insulation materials helping fix both environmental problems.

Objectives:

The objectives of this project are:

1. Develop eco-friendly bio-composite insulation panels using waste aquatic biomass primarily from water lily.
2. Optimize the desired characteristics through testing to improve performance

Summary of Progress:

The project on development of bio-composite insulation materials from waste aquatic biomass has made significant progress since initiation. The team has completed an extensive literature review (20+ papers, 2020-2025) to identify research gaps and conceptualize the experimental framework. Following this, raw material collection, pretreatment, and composite fabrication have been successfully carried out and left in the oven to dry for testing afterwards.

Work Completed:

1. Literature Review & Framework

- Surveyed more than 20 research papers on bio-based insulation, binder systems, flame retardancy, and moisture resistance
- Identified key research gaps: hybrid aquatic biomass systems, optimized natural binders, and moisture/fire performance improvement
- Designed composite variant matrix and finalized testing methodologies

2. Raw Material Collection & Pretreatment

- Collected water lily and water hyacinth from Dal Lake; washed thoroughly to remove dirt and debris
- Sun-dried the biomass, followed by oven drying to achieve consistent low-moisture content
- Ground the dried biomass using the mixer grinder (Dept. of Chemistry, under Prof. Tabassum Ara) to obtain fibrous particles
- Two particle size fractions obtained: coarse (>3 mm) and fine (1-1.5 mm)

3. Composite Fabrication - Four Samples Prepared

S.	Particle Size	Composition / Binder	ASTM E11 Mesh	Observation
S1 	Coarse >3 mm	Biomass only + 15% water	<No. 7 (>2.83 mm)	Poor mechanical stability; sample fractured on demoulding
S2 	Fine 1.0–1.5 mm	Biomass only + 15% water	No. 12-No. 18 (1.0-1.7 mm)	Better cohesion; intact panel formed
S3 	Fine 1.0–1.5 mm	Biomass:Starch = 90:10 (corn starch binder)	No. 12-No. 18 (1.0-1.7 mm)	Improved binding and surface finish with starch
S4 	Fine 1.0–1.5 mm	Biomass:Starch = 70:30 (corn starch binder)	No. 12-No. 18 (1.0-1.7 mm)	Compact, denser panel; highest binder fraction

4. Testing Results (completed)

Moisture Content - ASTM D4442 $[MC (\%) = (W^I - W^K) / W^I \times 100]$

S.	Initial Wt. (g)	Final Wt. (g)	Loss (g)	MC (%)	Remark
S1	18.1	16.3	1.8	9.94	Low MC - coarse, highly porous structure
S2	16.6	13.5	3.1	18.67	Moderate MC - fine particles, larger surface area
S3	16.3	12.4	3.9	23.93	Higher MC - corn starch retains moisture
S4	10.2	6.0	4.2	41.18	Highest MC - 30% starch binder strongly hygroscopic

Bulk Density - ASTM D1037 [ρ (g/cm³) = Oven-dry mass / Volume]

S.	Mass (g)	L×B×H (cm)	Vol. (cm ³)	ρ (g/cm ³)	Remark
S1	16.3	5×5×0.6	15.00	1.087	Highest density - coarse particles, compact fill
S2	13.5	5×5×0.6	15.00	0.900	Lower density - fine particles, water only binder
S3	12.4	5×5×0.5	12.50	0.992	Moderate - 10% starch increases compaction
S4	6.0	4.8×4.8×0.3	6.91	0.868	Lowest density - 30% starch, thinner panel

Water Absorption - ASTM D570, 2-hour soak [WA (%) = $(W^{Wet} - W^{dRy}) / W^{dRy} \times 100$]

S.	Dry Wt. (g)	Wet Wt. (g)	Gain (g)	WA (%)	Remark
S1	1.9	14.4	12.5	657.89	Very high - no binder, open porous coarse structure
S2	3.5	21.8	18.3	522.86	High - fine particles increase surface area for absorption
S3	2.8	17.0	14.2	507.14	High - corn starch itself is hydrophilic
S4	1.0	3.8	2.8	280.00	Relatively lower - higher binder fraction reduces open porosity

Pending Tests and Future Work Plan

Pending Test	Standard	Timeline	Next Steps After Testing
Mechanical Strength (Compressive/Flexural)	ASTM D695 / ASTM D790	Immediate after the Mid-term project presentation.	Optimize binder ratio; add flame retardants (Al(OH) ₃ , borax)
Flammability	ASTM D635 / UL-94 ISO 11925-2	Immediate after the Mid-term project presentation.	Develop FR variants
Thermal Conductivity	ASTM C518 / ASTM C177 ISO 22007-2	First week of May.	Target: 0.04-0.07 W/mK; compare vs. glass wool and PU foam